ZOO502 Mid Term Current Solve
Subjective

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Regards VUWAYS Team

Welcome To vuways Study Help

Function of olfactory epithelium?

The olfactory epithelium is a specialized epithelial tissue inside the nasal cavity that is involved in smell. In humans, it measures about 1 square centimetre (on each side) and lies on the roof of the nasal cavity about 7 cm above and behind the nostrils. The olfactory epithelium is the part of the olfactory system directly responsible for detecting odors. **(Internet)**

Hormones for salt and water maintenance?

Mineralocorticoids • Mineralocorticoids have effects on mineral metabolism. They act principally in maintaining salt and water balance in the body. The primary mineralocorticoid hormone is aldosterone **(PPTS)**

Three major hormones are involved in regulating sodium and water balance in the body at the level of the kidney. (1) ADH (antidiuretic hormone) from the posterior pituitary acts on the kidney to promote water reabsorption, thus preventing its loss in the urine. **(Internet)**

Stigma and ocelli?

Stigma: Eyespot or Stigma • Simplest photoreceptive structure. • Found in some protozoa e.g. euglena. • It is a bright red colored organelle & has carotenoid pigments. • It gives a sense of light and dark. • Helps in phototaxis.

Ocelli: Eyecups or Ocelli • Multicellular photo-receptive structures that consist of a cuplike depression containing photoreceptor cells. • Found in cnidarians and flatworms e.g. Planaria. • Cannot form image. • Provide the animal a sense of direction only.

Function of estrogen during pregnancy?

During pregnancy, estrogens exert mainly a proliferative function on reproductive and associated organs of the mother.

Androgen secreted from adrenal cortex?

Hormones of Adrenal Cortex • Adrenal cortex also secretes small amounts of sex hormones, especially androgenic hormones. • These androgens exhibit about the same effects in the body as the male sex hormone testosterone.

Role of Fibers of Zonula and Ciliary Muscles?

- Fibers of zonula held the lens in place. • They change the shape of lens by exerting an outward tension on perimeter.
- Attached with fibers of zonula are the ciliary muscles • They contract to pull fibers of zonula, which flatten the lens. This focuses distant objects on retina. • When they relax, lens becomes rounded and near objects are focused.

Sensory receptor?

A sensory receptor is a structure that reacts to a physical stimulus in the environment, whether internal or external. It is a sensory nerve ending that receives information and conducts a process of generating nerve impulses to be transmitted to the brain for interpretation and perception. **(Internet)**

cGMP and cAMP difference?

Production of cGMP is catalyzed by guanylate cyclase enzyme from GTP. • Guanylate cyclase occurs in two forms: one bound to the membrane and one free in cytoplasm. In contrast, adenylate cyclase is always bound to the membrane. Guanylate cyclase becomes active as the Ca2+ concentration is increased within the cell, while adenylate cyclase activity is increased when Ca2+ conc. is low. cGMP activates a specific protein kinase G instead of protein kinase A.

Glandular cell and hormones of adenopathy?

Somatotropes—produce growth hormone (GH) 2.Corticotropes—produce adrenocorticotropic hormone (ACTH) 3.Thyrotropes—produce thyroid-stimulating hormone (TSH) 4.Gonadotropes—produce gonadotropic hormones (luteinizing hormone, LH and follicle stimulating hormone, FSH) 5.Lactotropes—produce prolactin (PRL) **(PPTS)**

There is another type of gland called an exocrine gland (e.g. sweat glands, lymph nodes). These are not considered part of the endocrine system as they do not produce hormones and they release their product through a duct. ... These hormones affect many parts of the human body. **(Internet) Coding line? Line Coding? Lable Line Coding?**

Each receptor subtype for taste sensations is connected to a particular set of axons in the nerve. • In such an arrangement, information about one taste e.g. "sweetness" would be carried by some specific subset of axons. • Such a pattern is called labeled line coding.

Negative Feed Back?

Negative feedback is a reaction that causes a decrease in function. It occurs in response to some kind of stimulus. Often it causes the output of a system to be lessened; so, the feedback tends to stabilize the system. This can be referred to as homeostatis, as in biology, or equilibrium, as in mechanics. **(Internet)**

Sensory transduction?

Definition: The physical or chemical energy of stimulus is converted into electrical signals i.e nerve impulse by sensory receptor cells. This conversion is called sensory transduction. All sensory transduction systems contain related molecules and operate through similar cellular mechanisms. • They perform three basic functions: • Detection of stimulus • Amplification of stimulus • Encoding of stimulus

Color blindness?

The three types of opsins in color pigments are encoded by three genes. • Gene encoding the opsin in blue-cone pigment is located on an autosomal chromosome. • The genes for red and green-cone pigments are closely linked on the X chromosome. Color blindness is caused due to a mutation in one of the cone opsin genes that results in the absence of one type of pigmented cones. • A person missing a single type of color receptive cones is unable to distinguish some colors.

Name of Two stimuli?

External stimuli Stimuli of odor, touch, light, sound and gravitation • **Internal stimuli** Stimuli of pain, homeostatic imbalances and blood pressure

Gland and Secretion?

Exocrine **glands secrete** substances out of the body and Endocrine **glands secrete** substances into capillaries and blood vessels. **(Jo mujy samjh aaya)**

Photoreceptor?

A **photoreceptor** cell is a specialized type of neuroepithelial cell found in the retina that is capable of visual phototransduction. The great biological importance of **photoreceptors** is that they convert light (visible electromagnetic radiation) into signals that can stimulate biological processes. **(Internet)**

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Function of LH in male and Female?

In **females**, an acute rise of **LH** ("**LH** surge") triggers ovulation and development of the corpus luteum. In **males**, where **LH** had also been called interstitial cell–stimulating hormone (ICSH), it stimulates Leydig cell production of testosterone. It acts synergistically with FSH. **(Internet)**

Structure of Sarcoplasmic reticulum?

The **sarcoplasmic reticulum** (SR) is a membrane-bound **structure** found within muscle cells that is similar to the endoplasmic **reticulum** in other cells. ... Therefore, it is vital that calcium ion levels are controlled tightly, and can be released into the cell when necessary and then removed from the cell. **(Internet)**

Define sensory adaptation? Give an example

Sensory adaptation is the process in which changes in the sensitivity of sensory receptors occur in relation to the stimulus. All senses are believed to experience sensory adaptation. However, some experimental psychologists say that the sense of pain does not experience this phenomenon.

Name different types of cells of islets of Langerhans ?

Islets of Langerhans which are endocrine. Islets of Langerhans • The islets contain four types of cells: 1. Beta Cells • 2. Alpha Cells •3. Delta Cells •4. PP Cells •

What are photoreceptors?

Photoreceptors are the cells in the retina that respond to light. Their distinguishing feature is the presence of large amounts of tightly packed membrane that contains the photopigment rhodopsin or a related molecule.

Describe at least three developmental abnormalities that arises due to Growth Hormone?

GHD can be present from birth (congenital), resulting from genetic mutations or from structural defects in the brain. It can also be acquired later in life as a result of trauma, infection, radiation therapy, or tumor growth within the brain.

Define Range Fractionation? Give it's importance as well as two examples?

In a sense organ, sensory receptors are arranged in an order of increasing sensitivity to different range of intensities of the stimulus. • This hierarchical arrangement of receptors is known as range fractionation.

Importance of Range Fractionation • It results in increasing the range of a multi-neuronal sensory system than the range of any individual receptor. • It increases the overall precision of sense organ. • It enables the CNS to discriminate stimulus intensities.

Write a brief note on signaling system of cAMP?

Signaling System of cAMP: Many hormones use the Adenylate Cyclase–cAMP second messenger signaling system to stimulate their target tissues. • The hormone binds to the receptor which is coupled to a G protein. G protein stimulates adenylate cyclase in the membrane. Activated AC catalyzes the conversion of ATP into cAMP in cytoplasm. cAMP then activates an enzyme called cAMPdependent protein kinase. • Protein kinase phosphorylates specific proteins in the cell, triggering a cascade of biochemical reactions that ultimately lead to cell's response to the hormone

Type of and name of difference cilia?

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Axon and its important?

Axon. An axon or nerve fiber, is a long, slender projection of a nerve cell, or neuron, in vertebrates, that typically conducts electrical impulses known as action potentials away from the nerve cell body.

Inner ear parts?

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It consists of a bony labyrinth which is filled with perilymph and has an oval and a round window. • Oval window receives sound vibrations through stapes. • In the bony labyrinth lies the membranous labyrinth that is filled with endolymph.

Placental Hormone?

The placenta is an endocrine gland that is only present during pregnancy. In this lesson, you will learn about the hormones it produces, including human chorionic gonadotropin (hCG), progesterone, estrogen, and human placental lactogen (hPL).

Hedging cycle?

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Donnan equilibiruim?

Donnan equilibrium, or Gibbs–Donnan equilibrium) is a name for the behaviour of charged particles near a semipermeable membrane that sometimes fail to distribute evenly across the two sides of the membrane.

Cardic Muscle and Skeletal Mucsle?

Cardiac and skeletal muscle are both striated in appearance, while smooth muscle is not. ... Cardiac muscle is also an involuntary muscle but is more akin in structure to skeletal muscle, and is found only in the heart.

Exocrine Gland with example?

Exocrine glands are glands that produce and secrete substances onto an epithelial surface by way of a duct. Examples of exocrine glands include sweat, salivary, mammary, ceruminous, lacrimal, sebaceous, and mucous.

What are endocrine gland give their name?

Endocrine glands are glands of the endocrine system that secrete their products, hormones, directly into the blood rather than through a duct. The major glands of the endocrine system include the pineal gland, pituitary gland, pancreas, ovaries, testes, thyroid gland, parathyroid gland, hypothalamus and adrenal glands.

What are androgen?

An androgen is any natural or synthetic steroid hormone that regulates the development and maintenance of male characteristics in vertebrates by binding to androgen receptors.

Vestibular apparatus?

Sensory information about motion, equilibrium, and spatial orientation is provided by the vestibular apparatus, which in each ear includes the utricle, saccule, and three semicircular canals. The utricle and saccule detect gravity (information in a vertical orientation) and linear movement.

Structure Sarcomere?–

Sarcomere: Sarcomeres are composed of long, fibrous proteins as filaments that slide past each other when a muscle contracts or relaxes. Two of the important proteins are myosin, which forms the thick filament, and actin, which forms the thin filament. ... Other bands appear when the sarcomere is relaxed.

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Water Ion?

water is not an ionic compound,it is a polar compound (having partial positive charges on hydrogen atoms and partial negative charge on oxygen).due to partial charges water in certain respects behave like ionic compounds e.g it splits into psitive and negative ions particularly in solutions.Shortly water is a polar ...

Taster receptor?

A taste receptor is a type of receptor which facilitates the sensation of taste. These receptors are of four types. When food or other substances enter the mouth, molecules interact with saliva and are bound to taste receptors in the oral cavity and other locations.

Interjuction?

An interjection is a word or expression that occurs as an utterance on its own and expresses a spontaneous feeling or reaction. It is a diverse category, encompassing a number of different parts of speech, such as exclamations, curses, greetings, response particles, hesitation markers and other words.

Inner receptor?

Olfactory receptor, also called smell receptor, protein capable of binding odour molecules that plays a central role in the sense of smell (olfaction). ... The outer and inner ends of receptor proteins involved in smell are connected by a chain of amino acids. **(Internet)**

Action model of neurotransmitter?

Neurotransmitters are released into and diffuse across the synaptic cleft, where they bind to specific receptors on the membrane of the postsynaptic neuron. ... Ultimately it will create a new action potential at its axon hillock to release neurotransmitters and pass on the information to yet another neighboring neuron.

Heart conduct system?

The cardiac conduction system is a group of specialized cardiac muscle cells in the walls of the heart that send signals to the heart muscle causing it to contract. The main components of the cardiac conduction system are the SA node, AV node, bundle of His, bundle branches, and Purkinje fibers. **(Internet)**

Acclimation?

Acclimatization or acclimatisation is the process in which an individual organism adjusts to a change in its environment, allowing it to maintain performance across a range of environmental conditions. Acclimatization occurs in a short period of time, and within the organism's lifetime.

Pituratory Horme name?

The anterior pituitary gland secretes six hormones. In this lesson, you will learn the functions of the hormones: growth hormone (GH), prolactin (PRL), follicle-stimulating hormone (FSH), luteinizing hormone (LH), adrenocorticotropic hormone (ACTH), and thyroid-stimulating hormone (TSH).

Theme of Physiology?

Major Goals of Study of Physiology: • Explore the basic physiological processes • Their evolution by selective forces They provide insight about: • Patterns of physiological evolution • Adaptive value of physiological processes.

Five Major Themes: 1.Structure-function relationships 2.Adaptation, Acclimatization and Acclimation 3.Principle of homeostasis 4.Feedback control systems 5.Conformity and Regulation

Neurogenic pacemaker and myogenic pacemaker?

Neurogenic and myogenic pacemaker

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system 2, 3 or 4 chambered hearts are seen with muscular ventricles which pumps the blood in the body with pressure and hence heart are known as pressure pumps. specialised muscle cells within the heart. They are known as Myogenic hearts. which later on may be become myogenic or neurogenic. **(Internet)**

Phases of action potential? an action potential occurs when the membrane potential of a specific cell location rapidly rises and falls: this depolarisation then causes adjacent locations to similarly depolarize

Gland form of interior pituratory?

The anterior pituitary contains five types of endocrine cell, and they are defined by the hormones they secrete: somatotropes (GH); Lactotropes (PRL); gonadotropes (LH and FSH); corticotropes (ACTH) and thyrotropes (TSH).

Feedback mechanism and principle and function?

Feedback Mechanism: It is the general mechanism of nervous or hormonal control and regulation in Human. Feedback system consists of a cycle of events in which information about a change is fed back into the system so that the regulator (brain) can control the process.

Name of thyroid gland hormones?



Hormones and its function?

Hormones are special chemical messengers in the body that are created in the endocrine glands. These messengers control most major bodily functions, from simple basic needs like hunger to complex systems like reproduction, and even the emotions and mood.

The main function of endocrine glands is to secrete hormones directly into the bloodstream. Hormones are chemical substances that affect the activity of another part of the body (target site). In essence, hormones serve as messengers, controlling and coordinating activities throughout the body.

Thyroid gland gastric intestinal tract?

Thyroid hormones define basal metabolism throughout the body, particularly in the intestine and viscera. Gastrointestinal manifestations of dysthyroidism are numerous and involve all portions of the tract. ... The liver is the most affected organ in both hypo- and hyperthyroidism.

Disease due to hypothyroidism?

Hypothyroidism results when the thyroid gland fails to produce enough hormones. Hypothyroidism may be due to a number of factors, including: Autoimmune disease. The most common cause of hypothyroidism is an autoimmune disorder known as Hashimoto's thyroiditis.

Name of cell of eye which involved in vision?

Cone cells, or cones, are photoreceptor cells in the retinas of vertebrate eyes (e.g. the human eye). They respond differently to light of different wavelengths, and are thus responsible for color vision and function best in relatively bright light, as opposed to rod cells, which work better in dim light.

Function: Color vision

Location: Retina of mammals

Gonadorpin?

Gonadotropins: Luteinizing and Follicle Stimulating Hormones. Luteinizing hormone (LH) and follicle-stimulating hormone (FSH) are called gonadotropins because stimulate the gonads - in males, the testes, and in females, the ovaries. They are not necessary for life, but are essential for reproduction.

Define sensory adaptation? Give example?

Sensory adaptation is the process in which changes in the sensitivity of sensory receptors occur in relation to the stimulus. All senses are believed to experience sensory adaptation. However, some experimental psychologists say that the sense of pain does not experience this phenomenon.

What are photo receptor?

Special cells in the eye's retina that are responsible for converting light into signals that are sent to the brain. Photoreceptors give us our color vision and night vision. There are two types of photoreceptor cells: rods and cones. A number of eye problems can involve photoreceptor cells.

Define the taste buds and also taste receptor?

Taste buds contain the taste receptor cells, which are also known as gustatory cells. ... Via small openings in the tongue epithelium, called taste pores, parts of the food dissolved in saliva come into contact with the taste receptors.

Conductive system of heart and name if its component?

Heart has: • A specialized system for generating rhythmic electrical impulses. • A system to conduct these impulses rapidly throughout its muscles. • These impulses cause and control rhythmic contraction of cardiac chambers. Component: Atrial internodal pathways (conduct impulse from S-A node to the A-V node). 2.A-V node (conducts impulses into the ventricles). 3.A-V bundle (bundle of His), conducts impulses from atria into the ventricles. 4. Left & right branches of A-V bundle. 5.Purkinje fibers, branch off from bundle branches (conduct impulse to all parts of the ventricles).

Positive and negative feedback?

A positive feedback mechanism is the exact opposite of a negative feedback mechanism. With negative feedback, the output reduces the original effect of the stimulus. In a positive feedback system, the output enhances the original stimulus. ... Another good example of a positive feedback mechanism is blood clotting.

RT Mechanism?

Nehnam midillation such as Q?

Messine corporuls and pacciners corpuscles?

Neuriphrien and phuphrien?

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